

Malabsorption — Differential Lineup



1. Whipple — PAS macrophages

WHAT YOU SEE

First lineup man holding a pink-stained card 'WHIPPLE / PAS MACROPHAGES', far left.

WHAT IT ENCODES

Whipple disease = infection with Tropheryma whipplei; small-bowel biopsy shows PAS-positive foamy macrophages. Middle-aged white men. Classic tetrad weight loss, diarrhea, arthralgia, abdominal pain; arthralgia often precedes GI by years.

2. Whipple — CNS & cardiac

WHAT YOU SEE

Tags 'CNS', 'CARDIAC', 'SIX YEARS EARLY' around the first figure, upper-left.

WHAT IT ENCODES

Whipple has prominent extra-intestinal features: CNS (dementia, ophthalmoplegia, the pathognomonic oculomasticatory myorhythmia) and CARDIAC (culture-negative endocarditis). Arthralgias can precede diagnosis by ~6 years. CNS relapse is feared.

3. Whipple — long antibiotics

WHAT YOU SEE

A 'LONG ANTIBIOTICS' pill tag at the first man's feet, lower-left.

WHAT IT ENCODES

Treat Whipple with PROLONGED antibiotics: IV ceftriaxone (CNS-penetrating) 2 weeks, then oral TMP-SMX for ~1 year. Under-treatment risks CNS relapse. Lifelong follow-up.

4. Tropical sprue

WHAT YOU SEE

Second man in a tropical shirt holding 'TROPICS', center-left.

WHAT IT ENCODES

Tropical sprue = post-infectious malabsorption in/after travel to tropics, causing folate and B12 deficiency (megaloblastic anemia) and villous changes. Diffuse small-bowel involvement.

5. Sprue: antibiotics + folate

WHAT YOU SEE

'FOLATE' and 'B12' tags on the tropical man, center.

WHAT IT ENCODES

Treat tropical sprue with antibiotics (tetracycline ~3-6 months) PLUS folate (and B12). It responds to antibiotics — confirming an infectious etiology. Replete the deficiencies.

6. Gluten-free fails (X)

WHAT YOU SEE

A crossed-out bread loaf 'GLUTEN-FREE FAILS' at the tropical man's feet, lower-center-left.

WHAT IT ENCODES

A gluten-free diet does NOT help tropical sprue — that is celiac's treatment. Tropical sprue needs antibiotics + folate. Mixing up the two is the classic trap.

BOARD TRAP

"Treat tropical sprue with a gluten-free diet" → wrong; gluten-free is for celiac. Tropical sprue responds to antibiotics (tetracycline) plus folate/B12.

7. Pancreatic insufficiency

WHAT YOU SEE

Third man clutching abdomen with 'CHRONIC PANCREATITIS' pancreas image, center-right.

WHAT IT ENCODES

Pancreatic exocrine insufficiency (chronic pancreatitis, CF, cancer) -> maldigestion of fat -> steatorrhea, weight loss, fat-soluble vitamin deficiency. Note: in pure pancreatic insufficiency the bowel mucosa is normal so amino-acid/D-xylose absorption is intact.

8. Low fecal elastase + enzymes

WHAT YOU SEE

A gauge 'ELASTASE LOW' and 'ENZYME REPLACING' jar near the third man, center-right.

WHAT IT ENCODES

Diagnose with LOW fecal elastase-1 (and high fecal fat). Treat with pancreatic enzyme replacement therapy (lipase with meals) plus a PPI, plus fat-soluble vitamin repletion. Manage the underlying pancreatitis.

9. Steatorrhea

WHAT YOU SEE

Greasy stool puddle labeled 'STEATORRHEA' below the third man, lower-center.

WHAT IT ENCODES

Steatorrhea = bulky, greasy, foul, floating stools from fat malabsorption; quantified by 72h fecal fat or qualitative Sudan stain. Common to pancreatic insufficiency, bile-salt problems, and mucosal disease.

10. Vit A — night blindness

WHAT YOU SEE

Banner 'A — NIGHT BLINDNESS' across the top-left ruler header.

WHAT IT ENCODES

Fat-soluble vitamin A deficiency -> night blindness, xerophthalmia, Bitot spots. Replete vitamin A. Part of the ADEK malabsorption cluster.

11. Vit D — osteoporosis

WHAT YOU SEE

Banner 'D — OSTEOPOROSIS' along the top, center-left.

WHAT IT ENCODES

Vitamin D deficiency -> osteomalacia/osteoporosis, hypocalcemia, bone pain. Replete vitamin D + calcium. Check in any chronic malabsorption.

12. Vit E — ataxia/neuropathy

WHAT YOU SEE

Banner 'E — ATAXIA NEUROPATHY' along the top, center-right.

WHAT IT ENCODES

Vitamin E deficiency -> spinocerebellar ataxia, peripheral neuropathy, hemolysis. A neurologic mimic; replete vitamin E.

13. Vit K — high INR

WHAT YOU SEE

Banner 'K — HIGH INR' with a clock at top-right.

WHAT IT ENCODES

Vitamin K deficiency -> impaired factors II, VII, IX, X -> elevated INR and bleeding. Correct with vitamin K. A clue that fat-soluble vitamin absorption is impaired.

14. Ileal resection >100 cm

WHAT YOU SEE

Fourth man at a ruler 'UNDER 100 CM / OVER 100 CM' pouring fluid, right.

WHAT IT ENCODES

Terminal ileal resection >100 cm causes BILE-SALT depletion (liver can't keep up) -> fat malabsorption and steatorrhea. Manage with a LOW-FAT diet (+ fat-soluble vitamins, B12); here cholestyramine binds the already-scarce bile salts and WORSENS steatorrhea.

BOARD TRAP

"Give cholestyramine for >100 cm ileal resection" → wrong; with extensive resection bile-salt pool is depleted, so cholestyramine worsens fat malabsorption. Use a low-fat diet.

15. <100 cm → cholestyramine helps

WHAT YOU SEE

A 'CHOLESTYRAMINE' tag near the ruler, right side.

WHAT IT ENCODES

Limited ileal resection <100 cm causes bile-SALT (choleretic) diarrhea: unabsorbed bile acids irritate the colon, but the bile-salt pool is preserved. Cholestyramine binds the offending bile acids and HELPS. The length cutoff (100 cm) flips management.

16. Low-fat diet (X for <100cm context)

WHAT YOU SEE

A crossed-out 'LOW-FAT DIET' plate at the fourth man's feet, lower-right.

WHAT IT ENCODES

For the SHORT (<100 cm) resection with bile-salt diarrhea, the fix is cholestyramine (not a low-fat diet); the low-fat diet belongs to the >100 cm fat-malabsorption scenario. Matching the right intervention to the resection length is the tested point.

BOARD TRAP

"Low-fat diet for short (<100 cm) ileal resection" → wrong; that scenario is bile-salt diarrhea treated with cholestyramine. Low-fat diet is for >100 cm fat malabsorption.